

E-Commerce Business Intelligence & Analytics Pipeline

From Raw Data to Revenue-Saving Decisions

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UK Online Retail (2010-2011)

The Story Behind the Data

The Situation:

An online retailer had ~540,000 transactions, but they were sitting in spreadsheets, untouched. The CEO couldn't answer three simple questions:

The Questions:

1. Who are our most valuable customers, and are they leaving us?
2. Why is our customer retention falling year over year?
3. What products should we bundle to increase revenue?

The Goal:

Transform this messy data into a clear, actionable dashboard that answers these questions in seconds.

The Solution : How I Built It

**EXCEL Cleaning &
EDA**



**POSTGRESQL Views &
Logic**



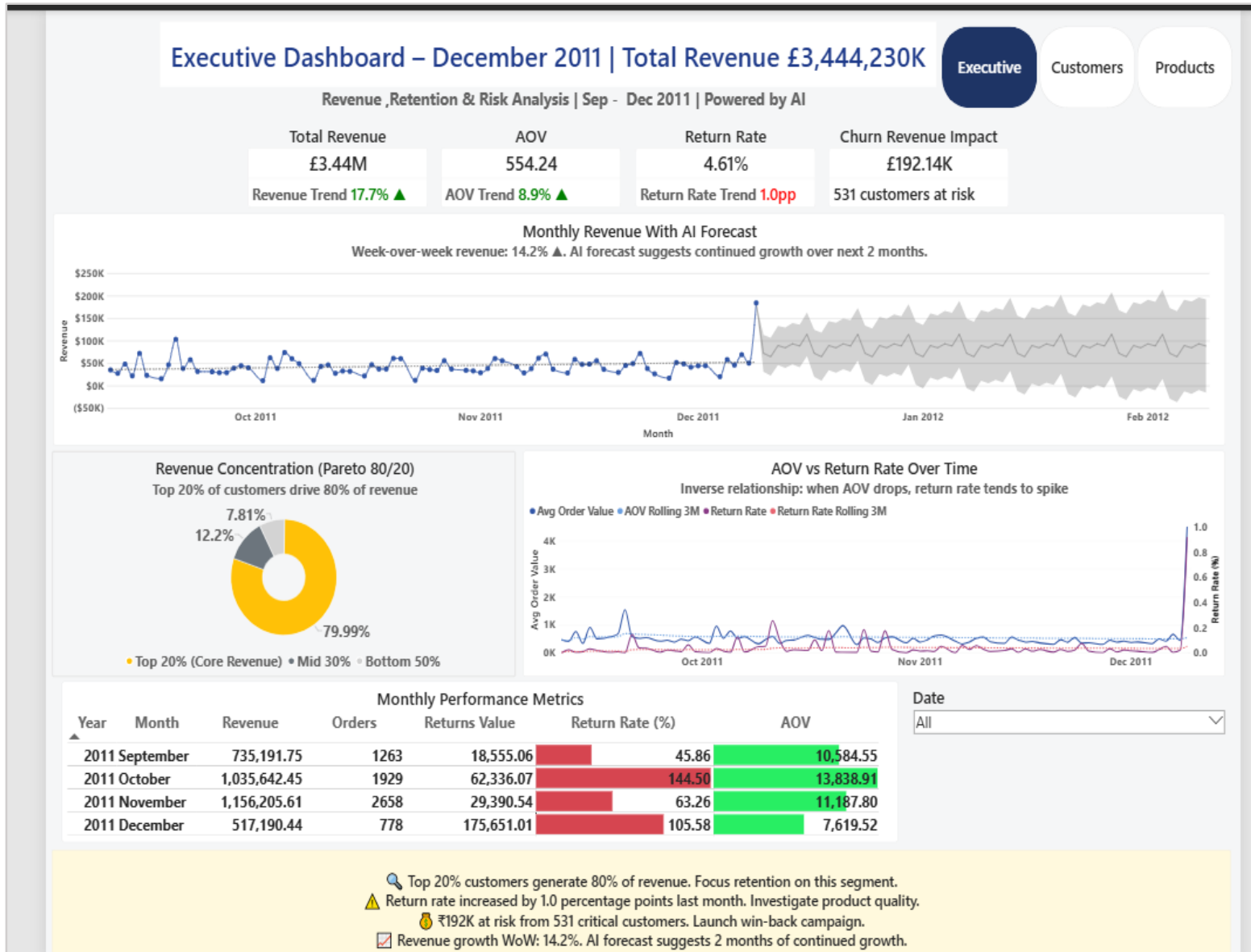
**POWER BI
DASHBOARD**

- ◆ Removed duplicates, handled missing IDs.
- ◆ Created 5 advanced SQL Views (CLV, Churn, Cohort, KPI, Affinity)
- ◆ Built 3 interactive Power BI pages with AI visuals.

💡 Why This Stack ?

Excel for quick exploration. SQL for heavy logic. Power BI for storytelling.

Executive Dashboard - Business Health at a Glance



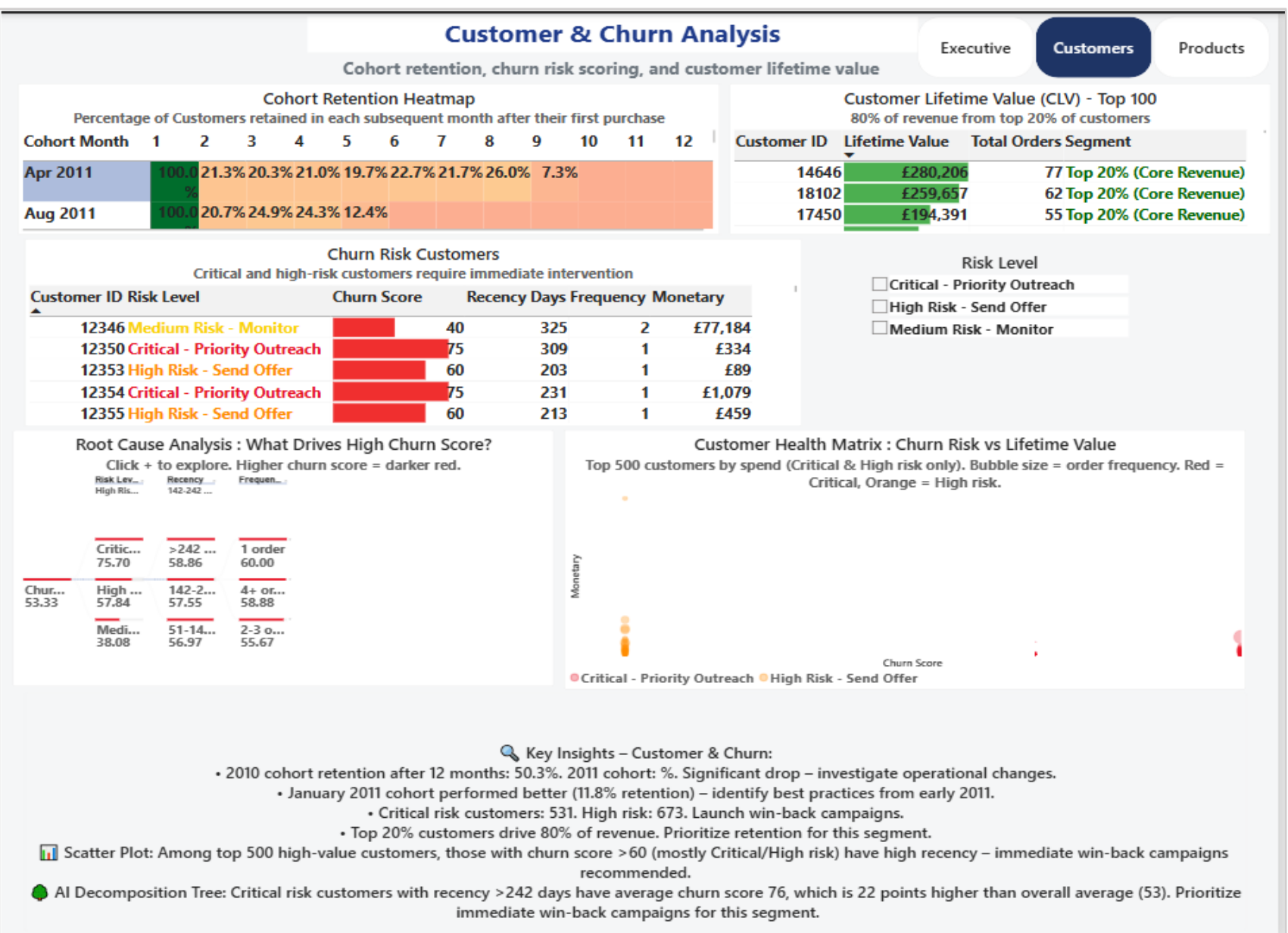
Key Metrics & Insights

- ✓ Revenue: £3.44M
- ✓ AOV: £554
- ✓ Return Rate: 4.61%
- ✓ Revenue at Risk: £192K
- ✓ Top 20% = 80% Revenue
- 📈 WoW Growth: +14.2%

My Take:

"Protect top 20% customers – they are the engine."

Customer & Churn Analysis – Who's Leaving ?



⚠ Critical Findings

▼ Retention Drop:
2010: 50% → 2011: 12%

● At-Risk Customers:
531 Critical + 673 High-Risk

AI Insight:
Critical + Recency > 242 days
= 76 churn score (avg 53)

💡 Action:

"Target 'Old & Critical' segment with personalized offers."

Product Affinity – What Sells Together ?

Product Affinity & Cross-Selling Analysis

Frequently bought together - prioritize bundles with high lift & confidence

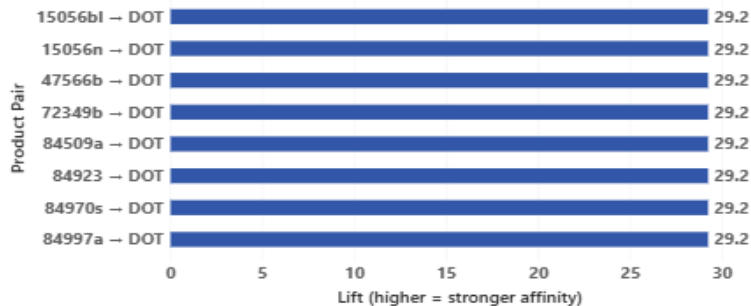
Executive

Customers

Products

Top Product Pairs by Lift (Affinity Score)

Only Pairs with lift >0 are shown. Lift > 1 indicates positive cross-sell potential



Max Lift

Strongest cross-sell score

29.24

Total Pairs

Filtered pairs (together_count >20, confidence >5%)

50

Detailed Product Affinity Matrix

Pairs sorted by lift (descending). Higher lift = stronger cross-sell opportunity

Product A	Product B	Together Count	Support (%)	Confidence (%)	Lift
15056bl	DOT	62	0.30	100.00	29.24
15056n	DOT	84	0.41	100.00	29.24
47566b	DOT	102	0.49	100.00	29.24
72349b	DOT	82	0.40	100.00	29.24
84509a	DOT	68	0.33	100.00	29.24
84923	DOT	98	0.47	100.00	29.24
84970s	DOT	66	0.32	100.00	29.24
84997a	DOT	65	0.31	100.00	29.24

Cross-Selling Insights:

- Highest lift (29.24) shared by 8 product pairs – all include product "DOT" paired with various others (84923, 15056bl, 15056n, etc.).
 - These pairs have 100% confidence: every time product A is bought, product B is also bought in the same transaction.
- Recommendation: Bundle "DOT" with any of these products and offer a 5-10% discount to increase average order value.
 - Total evaluated pairs: 50 (support >20, confidence >5%). Only 8 pairs show genuine affinity (lift > 1).
 - Use these high-lift pairs in email campaigns: "Customers who bought X also bought DOT".

What I Discovered

- ✓ 8 Pairs with Lift 29.24
- ✓ Product "DOT" in every high-lift pair
- ✓ Example:
 - 84923 → DOT
 - 15056bl → DOT
 - 47566b → DOT

Strategy:

"Bundle 'DOT' with top products. Offer 10% off to boost AOV."

Final Recommendations & Summary

✓ What I Would Do Next:

- ✦ VIP Loyalty Program – Protect the top 20% customers.
- ✦ Win-back Campaigns – Target 1,200+ at-risk customers.
- ✦ Bundle "DOT" – Launch cross-selling offers.
- ✦ Investigate 2011 Retention Drop – Fix declining loyalty.

🎯 Final Thought:

"This project proves that data isn't just numbers. It's a story about your customers. And when you listen to that story, you can save revenue, retain loyalty, and grow smarter."

🔗 Let's Connect:

LinkedIn: [Link] | GitHub: [Link]